ME 281 – Engineering Materials

Lecture 3: Structure of Crystalline Solids

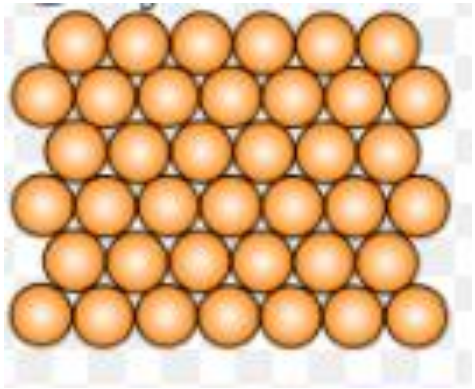
Learning Objectives

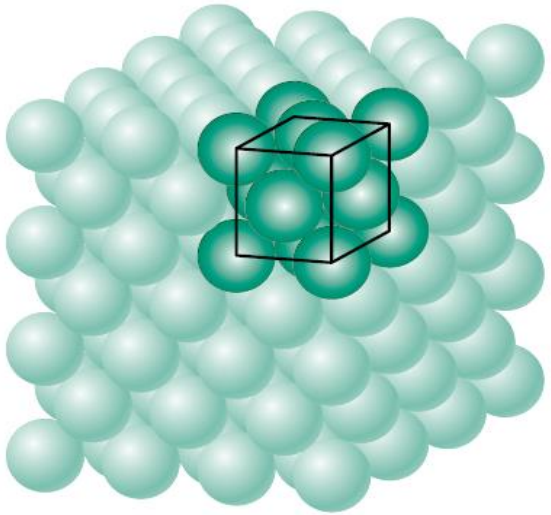
- Explain the common crystal structures in metals.
  - ✓ Face-centred cubic (FCC)
  - ✓ Body-centred cubic (BCC)
  - ✓ Hexagonal closed-packed (HCC)
- Understand the terms ***coordination number*** **(CN)** and ***atomic parking factor*** **(APF)**
- Explain types of solids (polymorphism, allotropy, polycrystalline and amorphous solids)
- Define Isotropy and Anisotropy

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Fundamental Concepts

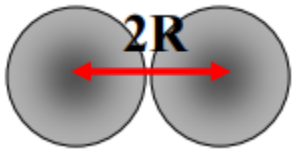
- Solid materials may be classified according to the regularity with which atoms or ions are arranged with respect to one another.
- A **crystalline** material is one in which the atoms are situated in a repeating or periodic array over large atomic distances; that is, long-range order exists, such that upon solidification, the atoms will position themselves in a repetitive three-dimensional pattern, in which each atom is bonded to its nearest-neighbor atoms.



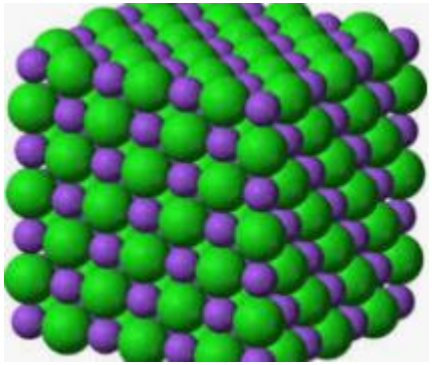


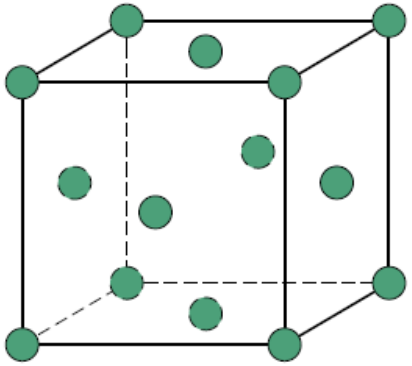
Fundamental Concepts

- Some of the properties of crystalline solids depend on the **crystal structure** of the material, the manner in which atoms, ions, or molecules are spatially arranged.
- There is an extremely large number of different crystal structures all having long-range atomic order; these vary from relatively simple structures for metals to exceedingly complex ones, as displayed by some of the ceramic and polymeric materials.
- When describing crystalline structures, atoms (or ions) are thought of as being **solid spheres** having well-defined diameters. This is termed the ***atomic hard-sphere model*** in which spheres representing nearest-neighbour atoms touch one another.



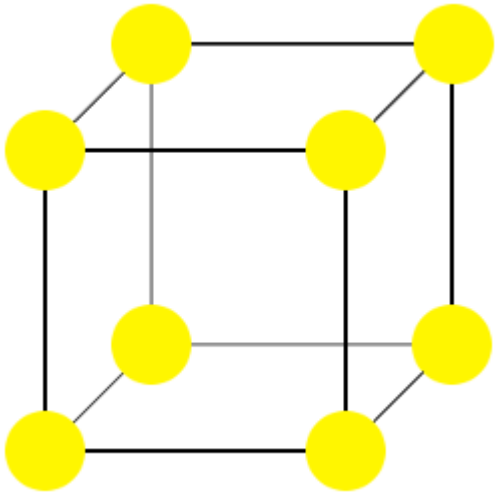
Hard-sphere model

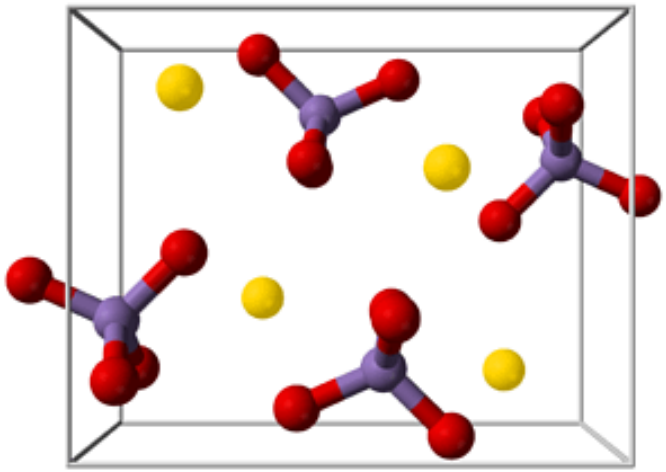


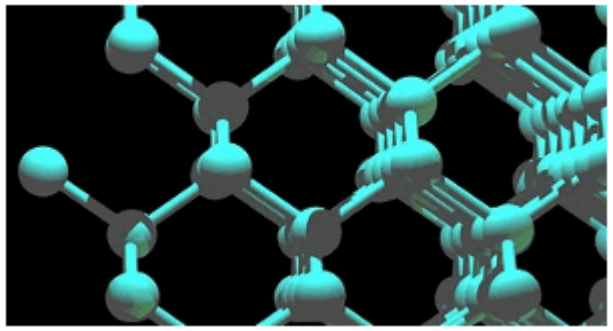


Fundamental Concept

- Sometimes the term **lattice** is used in the context of crystal structures; in this sense *lattice* means a ***three-dimensional array*** of points coinciding with atom positions (or sphere centres).

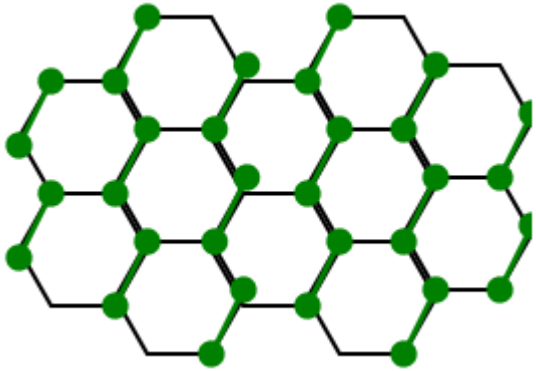




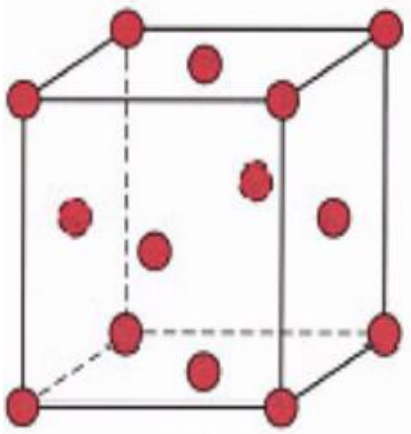


Unit Cells

- The **unit cell** is a structural unit or building block that can describe the crystal structure. Repetition of the unit cell generates the entire crystal.
- ***Example:*** 2D honeycomb net can be represented by translation of two adjacent atoms that form a unit cell for this 2D crystalline structure.
- Unit cells for most crystal structures are parallelepipeds or prisms having three sets of parallel faces.



2D honeycomb unit cell



Parallelepiped unit cell

Metallic Crystal Structure

- Metals are usually **(poly)crystalline**; although formation of **amorphous metals** is possible by rapid cooling.
- The atomic bonding in this group of materials is metallic and thus non-directional in nature.
- The most common types of unit cells are
  - faced-centred cubic (FCC)
  - body-centred cubic (BCC)
  - hexagonal close-packed (HCP).

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Characteristics of Crystal Structures

- Two important characteristics of a crystal structure are the **coordination number** and the **atomic packing factor (APF).**
- For metals, each atom has the same number of nearest-neighbour or touching atoms, which is the **coordination number (CN).**
- The atomic parking factor (APF) is the ratio of total atomic volume to the unit cell volume. This is expressed as

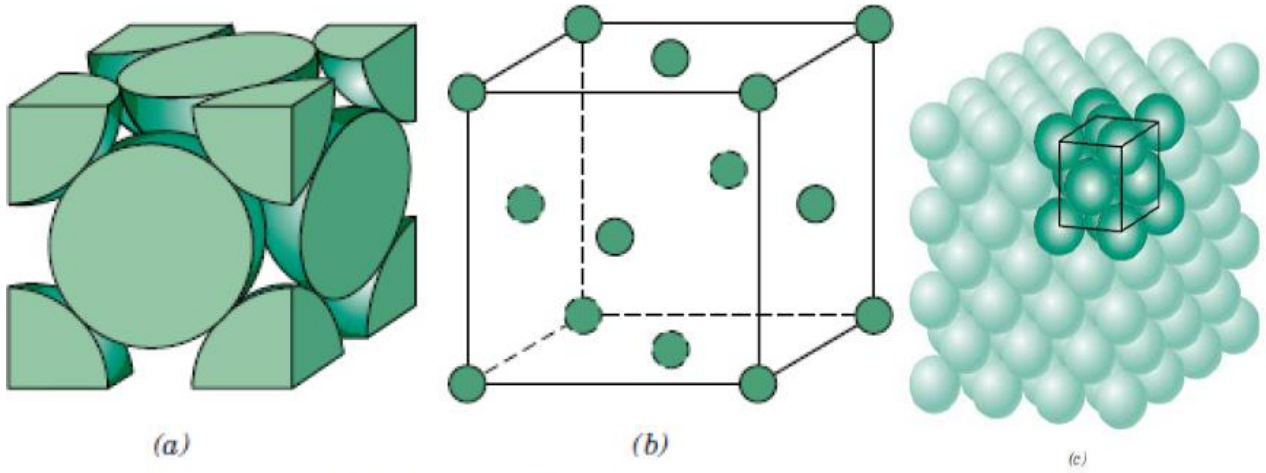
$$APF = \frac{Total\ atomic\ volume}{Unit\ cell\ volume} = \frac{N_{atoms\ per\ unit\ cell} \times V_{atom}}{V_{unit\ cell}} = \frac{V_s}{V_c}$$

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Faced-centred Cubic

- Atoms are located at each of the corners and on the centres of all the faces of cubic unit cell
- Cu, Al, Ag, Au have this crystal structure
- These spheres or ion cores touch one another across a face diagonal
- The cube edge length *a* and the atomic radius *R* are related through

$$a = 2R\sqrt{2}$$



For the face-centered cubic crystal structure, (*a*) a hard sphere unit cell representation, (*b*) a reduced-sphere unit cell, and (*c*) an aggregate of many atoms.

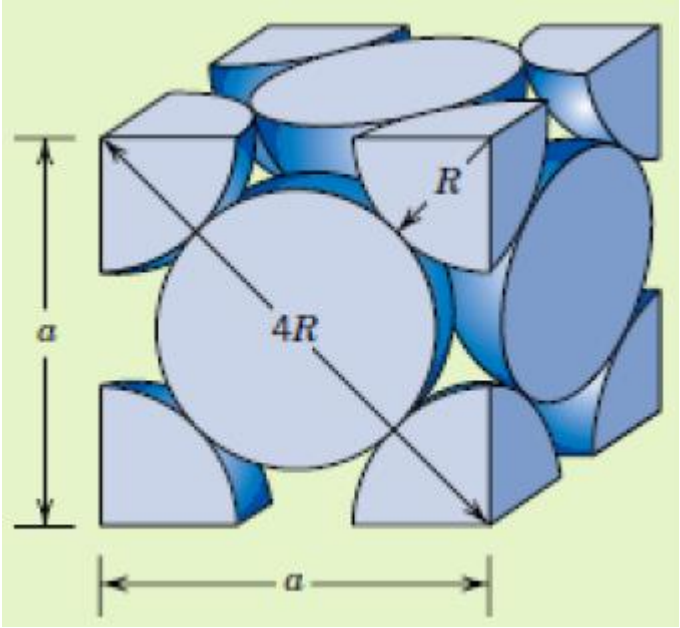
Face-centred Cubic

- In the FCC unit cell as illustrated, the atoms touch one another across a face-diagonal the length of which is 4*R*. Because the unit cell is a cube, its volume is a^3, where *a,* is the cell edge length.
- From the right angle triangle on the face, the edge length, a, is determined as,

$$a^2+a^2=(4R)^2$$

$$\Rightarrow a^2=1/2\times16R^2=8R^2$$

$$\Rightarrow \boldsymbol{a = 2R\sqrt{2}}$$



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Face-centred Cubic

- For the FCC crystal structure, each corner atom is shared among eight unit cells, whereas a face-centred atom belongs to only two.
- **Number of atoms per unit cell, n = 4**. (For an atom that is shared with m adjacent unit cells, we only count a fraction of the atom, 1/m). In FCC unit cell we have:

  6 face atoms shared by two cells: $6 \times 1/2 = 3$

  8 corner atoms shared by eight cells: $8 \times 1/8 = 1$
- The number of closest neighbours to which an atom is bonded = number of touching atoms. Thus coordination number, **CN = 12.**
- Atomic packing factor (APF) is 0.74
- APF for FCC is the maximum packing possible for spheres all having the same diameter.

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Computation of the Atomic Packing Factor for FCC

$$APF = \frac{Total\ atomic\ volume}{Unit\ cell\ volume} = \frac{N_{atoms\ per\ unit\ cell} \times V_{atom}}{V_{unit\ cell}} = \frac{V_s}{V_c}$$

Both the total atom and unit cell volumes may be calculated in terms of the atomic radius R. The volume for a sphere is $\frac{4}{3}\pi R^3$, and because there are four atoms per FCC unit cell, the total FCC atom (or sphere) volume is

$$V_s = (4) \times \frac{4}{3}\pi R^3 = \frac{16}{3}\pi R^3$$

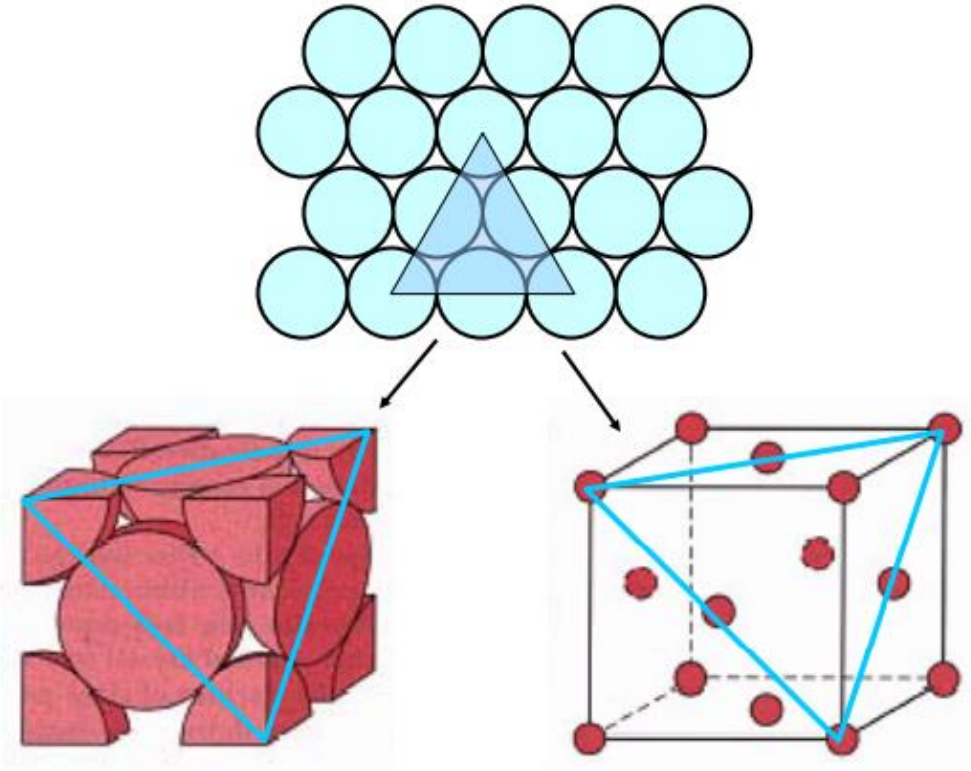
The total unit cell volume is $V_C = 16R^3\sqrt{2}$

Therefore, the atomic packing factor is

$$APF = \frac{V_s}{V_c} = \frac{\frac{16}{3}\pi R^3}{16R^3\sqrt{2}} = 0.74$$

Face-centred Cubic

- FCC is a close-packed crystal structure.
- FCC can be represented by a stack of close-packed planes (planes with highest density of atoms).

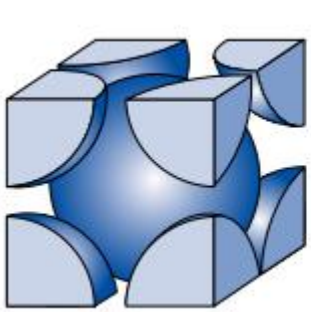
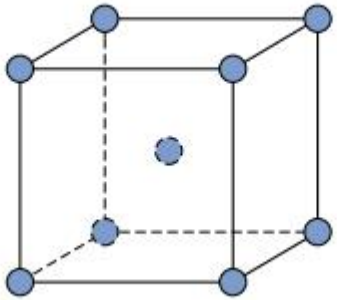
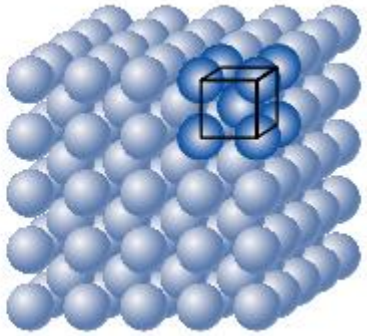


Body-centred Cubic

- Has a cubic unit cell with atoms located at all eight corners and a single atom at the cube centre.
- The centre and corner atoms touch one another along cube diagonals, and unit cell length, a, and atomic radius R, are related by

$$a = \frac{4R}{\sqrt{3}}$$

- Chromium, iron, tungsten, Molybdenum, Tantalum etc. exhibit a BCC structure.

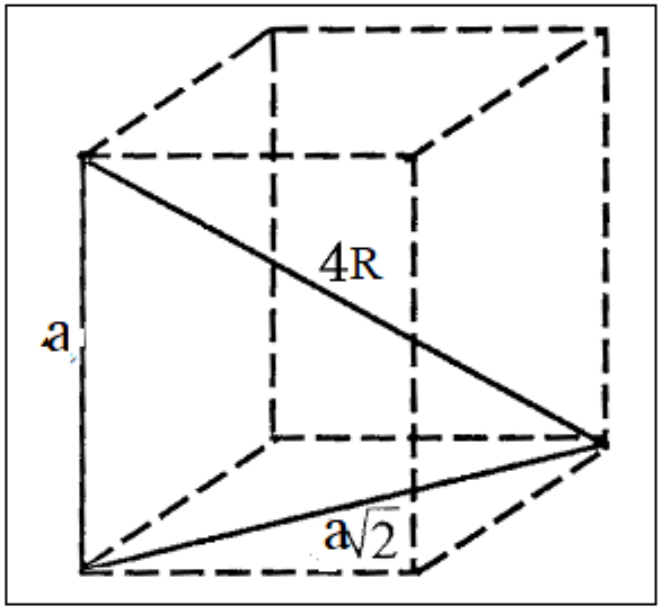




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Think-Pair-Share

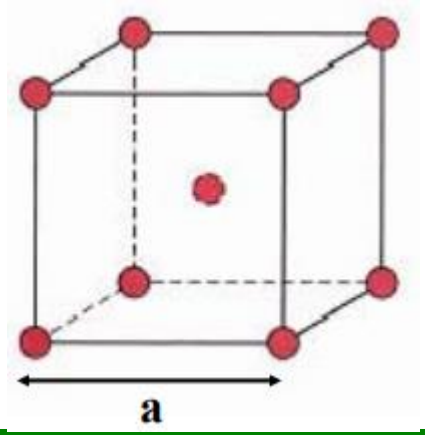
- Show that the edge length and atomic radius of BCC unit cell is related by

$$a = \frac{4R}{\sqrt{3}}$$



Body-centred Cubic

- The coordination number, CN = 8, lower than that for FCC.
- Number of atoms per unit cell, n = 2; thus,
  - ✓ centre atom (1) shared by no other cells: 1 x 1 = 1
  - ✓ 8 corner atoms shared by eight cells: 8 x 1/8 = 1
- Atomic packing factor, APF = 0.68.
- Corner and centre atoms are equivalent.
- Mechanical properties of metals with BCC structure are high yield strength, high young modulus, high hardness, high tensile strength and limited ability to forming.



Think-Pair-Share

- Show that the APF for BCC is 0.68

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Trial Example

The edge length of the unit cell of Ta, is 330.6 pm; the unit cell is body-centered cubic. Tantalum has a density of 16.69 g/cm^3.

(a) calculate the mass of a tantalum atom.
(b) Calculate the atomic weight of tantalum in g/mol.
(c) Calculate the radius of the atoms in the unit cell in pm

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Solution

1) Convert pm to cm:

$$330.6\,pm \times 1\,cm/1010\,pm = 330.6 \times 10^{-}10\,cm$$
$$= 3.306 \times 10^{-8}\,cm$$

2) Calculate the volume of the unit cell:

$$(3.306 \times 10^{-8}\,cm)3 = 3.6133 \times 10^{-}23\,cm^3$$

3) Calculate mass of the 2 tantalum atoms in the body-centered cubic unit cell:

$$(16.69\,g/cm^3)\,(3.6133 \times 10^{-23}\,cm^3)$$
$$= 6.0307 \times 10^{-22}\,g$$

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Solution Cont.

4) The mass of one atom of Ta:

$$6.0307 \times 10^{-22}\, g \,/\, 2 = 3.015 \times 10^{-}22\, g$$

5) The atomic weight of Ta in g/mol:

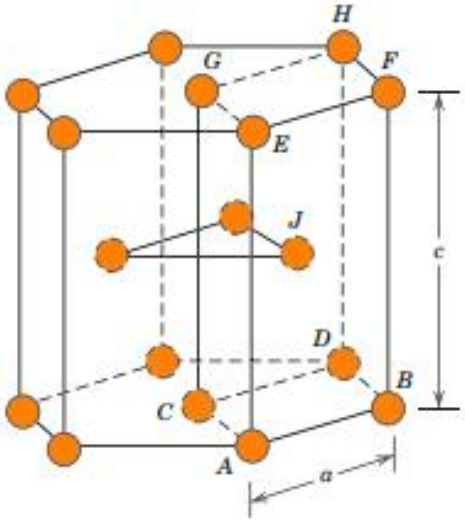
$$(3.015 \times 10^{-22}\, g)\,(6.022 \times 1023\, mol^{-}1)$$
$$= 181.6\, g/mol$$

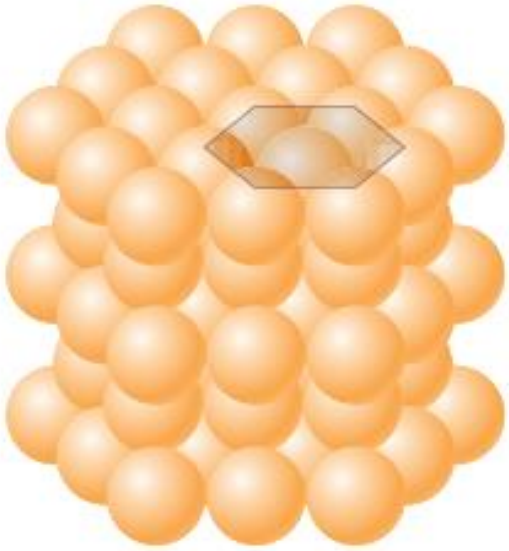
6) Radius of the atoms in the unit cell?

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Hexagonal Close-Packed (HCP)

- Some metals have crystal structures with unit cells which are not cubic in symmetry but rather hexagonal.
- Unit cell has two lattice parameters a and c. Ideal ratio c/a = 1.633.
- The coordination number, CN = 12 (same as in FCC)





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Hexagonal Close-Packed (HCP)

- Number of atoms per unit cell, n = 6.
  - ✓ 3 mid-plane atoms shared by no other cells: 3 x 1 = 3
  - ✓ 12 hexagonal corner atoms shared by 6 cells: 12 x 1/6 = 2
  - ✓ 2 top/bottom plane centre atoms shared by 2 cells: 2 x 1/2 = 1
- Atomic packing factor, APF = 0.74 (same as in FCC)
- $V_c = 6R^2c\sqrt{3}$ and a=2R
- All atoms are equivalent.
- Metals with HCP structure include cadmium, magnesium, titanium, zinc etc.
- Their mechanical properties are brittle, low yield strength and inability to foaming.

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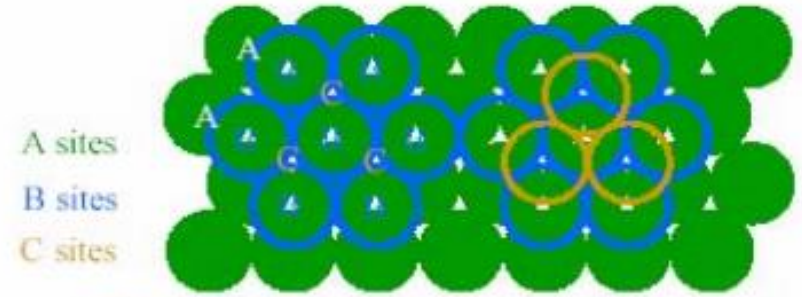
Close-packed Structures (FCC and HCP)

- Both FCC and HCP crystal structures have atomic packing factors of 0.74 (maximum possible value).
- Both FCC and HCP crystal structures may be generated by the stacking of close-packed planes.
- The difference between the two structures is in the stacking sequence.
- HCP has ABABA stacking sequence while FCC has ABCABC stacking sequence.

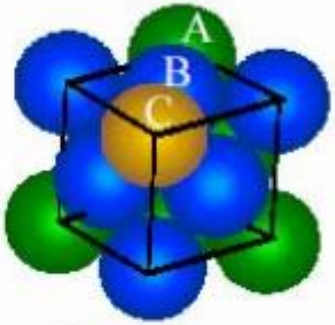
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FCC Stacking Sequence

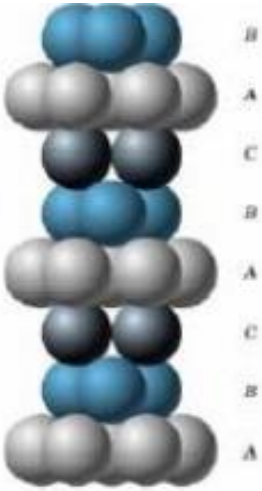
- ABCABC... Stacking Sequence
- 2D Projection

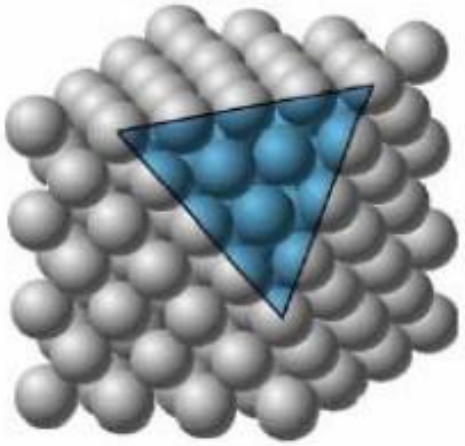






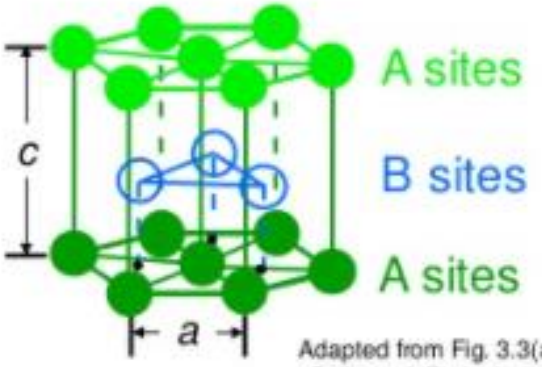
- FCC Unit Cell





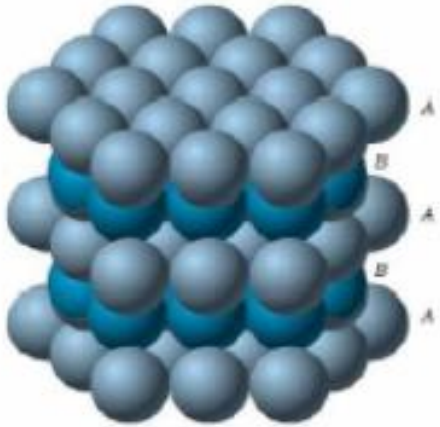
HCP Stacking Sequence

- ABAB... Stacking Sequence
- 3D Projection
- 2D Projection



Adapted from Fig. 3.3(a), *Callister & Rethwisch 3e.*





Density Calculations

- The crystal structure of a metallic solid permits computation of its theoretical density using the expression below:

$$\rho = \frac{nA}{V_c \times N_A}$$

Where

n is the number of atoms associated with each unit cell

A is the atomic weight

V_C is the volume of the unit cell and

N_A is the Avogadro's number.

- NB: Atoms in the unit cell, n = 2 (BCC); 4 (FCC); 6 (HCP)
- The volume of the cell, Vc = a^3 (FCC and BCC)

where a = $2R\sqrt{2}$ (FCC); a = $4R/\sqrt{3}$ (BCC) and R is the atomic radius

Worked Example

Copper has an atomic radius of 0.128 nm, an FCC crystal structure, and an atomic weight of 63.5 g/mol. Compute its theoretical density and compare answer with its measured density.

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Solution

- The crystal structure is FCC. Therefore, number of atoms per unit cell, n, is 4. Also, atomic weight A of copper (Cu) is given as 63.5 g/mol. The unit cell volume V_c for FCC is $16\,R^3\sqrt{2}$ where R, the atomic radius, is 0.128 nm. Substituting the parameters into the density equation gives

$$\rho = \frac{n\,A}{V_c\,N_A} = \frac{n \times A_{Cu}}{16\,R^3\,\sqrt{2}\,N_A}$$

$$\rho = \frac{\left(4\,\frac{atoms}{unit\ cell}\right) \times \left(63.5\,\frac{g}{mol}\right)}{\left[16\sqrt{2}\,(1.28 \times 10^{-8})^3\,\frac{cm^3}{unit\ cell}\right] \times \left(6.022 \times 10^{23}\,\frac{atoms}{mol}\right)} = 8.89\ g/cm^3$$

- The density of copper in literature is 8.94 g/cm3 , which is very close to the calculated value obtained.

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Think-Pair-Share

Zirconium has an HCP crystal structure and a density of 6.51 g/cm^3.

(a) What is the volume of its unit cell in cubic meters?

(b) If the c/a ratio is 1.593, compute the values of c and a.

[Take atomic weight of Zr =91.22g/mol]

Solution

(a) The volume of the Zr unit cell may be computed using

$$V_c = \frac{nAZ_r}{\rho N_A}$$

Now, for HCP, n = 6 atoms/unit cell, and for Zr, A = 91.22 g/mol. Thus,

$$V_c = \frac{(6 \times 9.122)}{6.51 \times 6.022 \times 10^{23}}$$

$$= 1.396 \times 10^{-22}\ cm^3/\text{unit cell}$$

$$= 1.396 \times 10^{-28}\ m^3/\text{unit cell}$$

Solution cont.

For HCP $V_c = 6R^2c\sqrt{3}$

But, since a = 2R, (i.e., R = a/2) then

$$V_c = 6\left(\frac{a}{2}\right)^2 c\sqrt{3} = \frac{3\sqrt{3}a^2c}{2}$$

but, since c = 1.593a

$$V_c = \frac{3\sqrt{3}(1.593)a^3}{2} = 1.396 \times 10^{-22}\ cm^3/\text{unit cell}$$

Solving for a

$$a = \left[\frac{(2)(1.396 \times 10^{-22})}{(3\sqrt{3})\times(1.593)}\right]^{1/3} = 3.23 \times 10^{-8} = 0.323\ \text{nm}$$

c = 1.593a = (1.593)(0.323 nm) = 0.515 nm

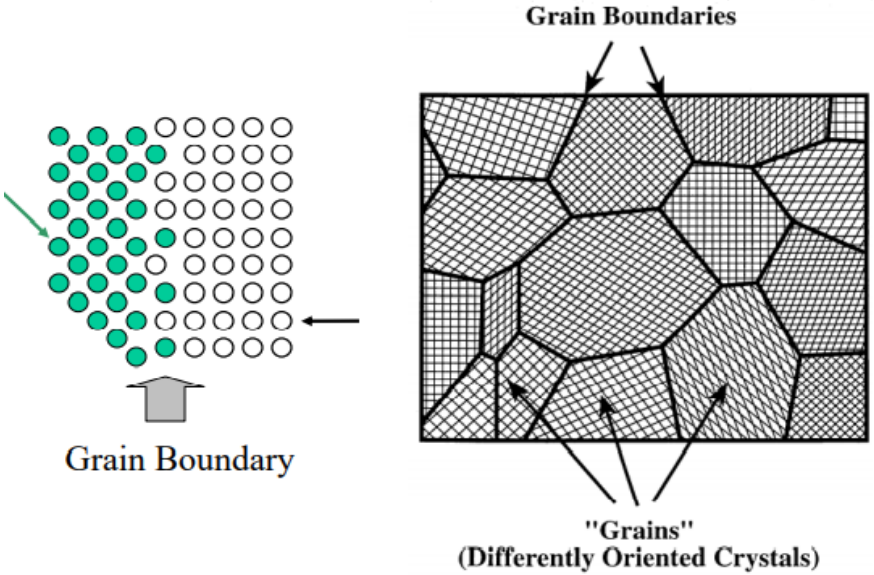
Polymorphism and Allotropy

- Some materials may exist in more than one crystal structure, this is called **polymorphism**.
- If the material is an elemental solid, it is called **allotropy**.
- The prevailing crystal structure depends on both the temperature and the external pressure.
- An example of allotropy is carbon, which can exist as diamond and graphite.
- Graphite is the stable polymorph at ambient conditions, whereas diamond is formed at extremely high pressures.
- Also, pure iron has a BCC crystal structure at room temperature, which changes to FCC iron at 912°C(1674°F).

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Crystal Systems

- **Single crystal**: atoms are in a repeating or periodic array over the entire extent of the material
- **Polycrystalline material**: comprised of many small crystals or grains.
- The grains have different crystallographic orientation. There exist atomic mismatch within the regions where grains meet. These regions are called grain boundaries.



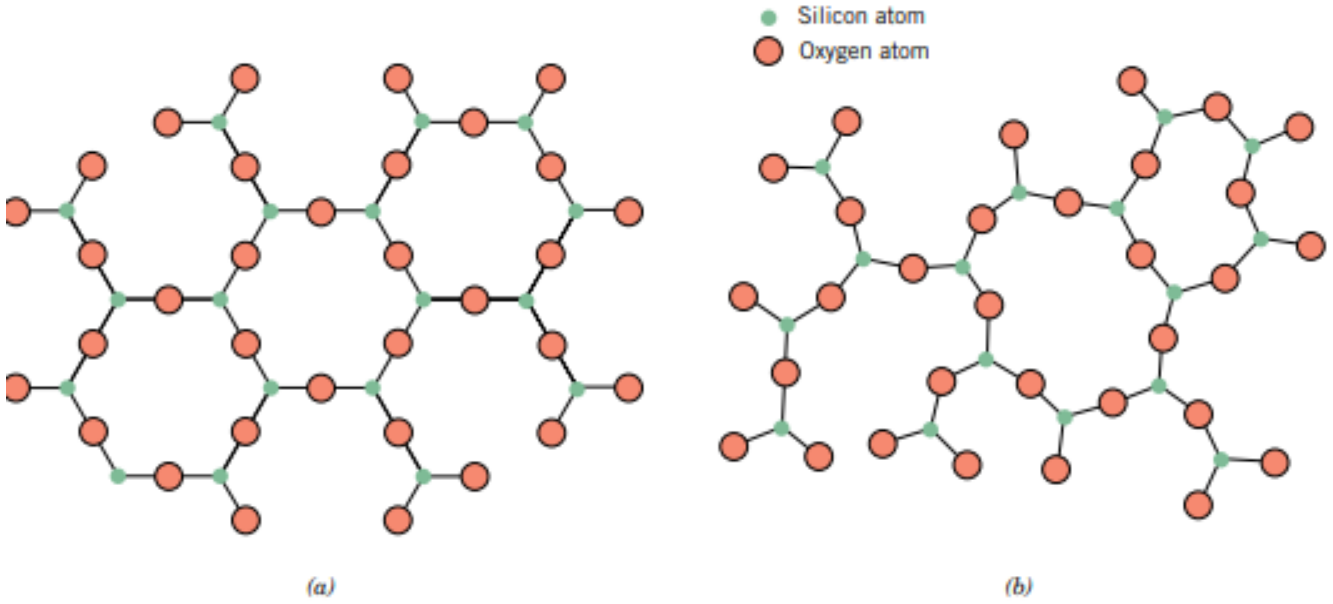
Anisotropy

- Different directions in a crystal have different packing.
- For instance, atoms along the edge of FCC unit cell are more separated than along the face diagonal.
- This causes **anisotropy** in the properties of crystals, for instance, the deformation depends on the direction in which a stress is applied.
- In some polycrystalline materials, grain orientations are random, so bulk material properties are **isotropic.**
- Some polycrystalline materials have grains with preferred orientations (texture), so properties are dominated by those relevant to the texture orientation and the material exhibits anisotropic properties.

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Non-Crystalline (Amorphous) Solids

- In amorphous solids, there is no long-range order.
- But amorphous does not mean random, in many cases there is some form of short-range order.



Two-dimensional schemes of the structure of (*a*) crystalline silicon dioxide and (*b*) noncrystalline silicon dioxide.

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Assignment

1. For an HCP, show that the ideal ratio c/a = 1.633.
2. Show that the atomic packing factor for HCP is 0.74.

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Thank You